Resultant

The algorithm to calculate gcd of two polynomials is useful. However, it is hampered by the fact that we need the coefficients to be in a field. There are many situations when we need to determine the existence of common factors when the coefficients are in a domain. For this, we introduce the resultant of two polynomials with coefficients in a domain Z.

Given a polynomial $\mathbf{a}$ of degree at most d and a polynomial $\mathbf{b}$ of degree at most e, we define a map

$$Z[X]_{<e} \oplus Z[X]_{<d} \to Z[x]_{<d+e} \text{ by } (\mathbf{p}, \mathbf{q}) \mapsto \mathbf{a} \cdot \mathbf{p} + \mathbf{b} \cdot \mathbf{q}$$

Here, for a positive integer r, we use $Z[X]_{<r}$ to denote the set of polynomials of degree less than r.

Using the basis $\{1, X, \ldots, X^{e-1}, 1, X, \ldots, X^{d-1}\}$ on the left-hand side we see that this linear map is given by

$$\{1, X, \ldots, X^{e-1}, 1, X, \ldots, X^{d-1}\} \mapsto \{\mathbf{a}, X\mathbf{a}, \ldots, X^{e-1}\mathbf{a}, \mathbf{b}, X\mathbf{b}, \ldots, X^{d-1}\mathbf{b}\}$$

Expressing this in terms of the basis $\{1, X, \ldots, X^{e+d-1}\}$ on the right-hand side, we obtain the *Sylvester matrix* of size $(d+e) \times (d+e)$.

$$\mathrm{Syl}_{d,e}(\mathbf{a}, \mathbf{b}) = \begin{pmatrix} a_0 & 0 & \ldots & 0 & 0 & b_0 & 0 & \ldots & 0 & 0 \\ a_1 & a_0 & \ldots & 0 & 0 & b_1 & b_0 & \ldots & 0 & 0 \\ a_2 & a_1 & \ldots & 0 & 0 & b_2 & b_1 & \ldots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & a_d & a_{d-1} & 0 & 0 & \ldots & b_e & b_{e-1} \\ 0 & 0 & \ldots & 0 & a_d & 0 & 0 & \ldots & 0 & b_e \end{pmatrix}$$

Here, the columns of a_i's occur e-times, each time downshifted once, this makes e column vectors of length $d+e$; the columns of b_j's occur d times, each time downshifted once, this meakes d column vectors of length $d+e$.

The *resultant* of size (d, e) of $(\mathbf{a}, \mathbf{b})$ is defined as the determinant of this matrix and is denoted by $\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b})$.

Some simplications

Vanishing. If the degree of $\mathbf{a} < d$ and the degree of $\mathbf{b} < e$, then the last row consists of 0's and thus the resultant is 0.

Thus, we are *primarily* interested in the case when $\mathbf{a}$ has degree d and $\mathbf{b}$ has degree e. However, the other cases may also occur in the simplification steps below.

Interchange. If we interchange $\mathbf{a}$ with $\mathbf{b}$ and d with e, we get a change of sign which can be seen to be $(-1)^{de}$.

$$\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) = (-1)^{de} \mathrm{Res}_{e,d}(\mathbf{b}, \mathbf{a})$$

Constant polynomials. Assume that $d = 0$ and $\mathbf{a} = a$ is a constant polynomial.

In this case, we are calculating $\mathrm{Res}_{0,e}(a, \mathbf{b})$. The Sylvester matrix in this case is the diagonal matrix of size $e \times e$ with a in all the diagonal entries. Thus we obtain $\mathrm{Res}_{0,e}(a, \mathbf{b}) = a^e$.

Column operations. Suppose that $k \leq d - e$ and c is some constant.

The matrix $\mathrm{Syl}_{d,e}(\mathbf{a} + cX^k \cdot \mathbf{b}, \mathbf{b})$ is obtained from the matrix $\mathrm{Syl}_{d,e}(\mathbf{a}, \mathbf{b})$ by multiplying various b-columns by c and adding that to a suitable a-column. In other words, this matrix is obtained from the other by elementary column opeations. Thus, the determinant does not change.

$$\mathrm{Res}_{d,e}(\mathbf{a} + cX^k \cdot \mathbf{b}, \mathbf{b}) = \mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b})$$

Reduction of d. When $\mathbf{a}$ has degree $\delta < d$, we note that the last $d - \delta$ rows of $\mathrm{Syl}_{d,e}(\mathbf{a}, \mathbf{b})$ are upper triangular with diagonal entry b_e. This shows that $\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) = b_e^{d-\delta}\mathrm{Res}_{\delta,e}(\mathbf{a}, \mathbf{b})$.

Simplification

Give that $\mathbf{a}$ and $\mathbf{b}$ are polynomials with coefficients in Z. Suppose that $Z \subset Z'$ is a subring of a domain Z'. Then we can think of $\mathrm{Syl}_{d,e}(\mathbf{a}, \mathbf{b})$ as a matrix with entries in Z' also. However, since its entries are actually in Z, the determinant $\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b})$ is in Z! Thus, we can perform *calculations* in a bigger domain Z' and be assured that the answer we obtain is still in Z.

We now put Z inside a field, such as $Q(Z)$, so that we can perform division of $\mathbf{a}$ by $\mathbf{b}$ when $\mathbf{b}$ is non-zero and its degree is less than that of $\mathbf{a}$.

We use long division to write $\mathbf{a} = \mathbf{d} \cdot \mathbf{b} + \mathbf{c}$ with the degree of $\mathbf{c}$ less than the degree of $\mathbf{b}$ and $\mathbf{a}$.

By repeated application of column operations we see that

$$\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) = \mathrm{Res}_{d,e}(\mathbf{c} + \mathbf{d} \cdot \mathbf{b}, \mathbf{b}) = \mathrm{Res}_{d,e}(\mathbf{c}, \mathbf{b})$$

Note that the degree f of $\mathbf{c}$ is less than d. Thus, we can use the reduction of d argument to write

$$\mathrm{Res}_{d,e}(\mathbf{c}, \mathbf{b}) = b_e^{d-f}\mathrm{Res}_{f,e}(\mathbf{c}, \mathbf{b})$$

We can now use the interchange to write this as $(-1)^{fe}\mathrm{Res}_{e,f}(\mathbf{b}, \mathbf{c})$ and repeat the process. (Note the similarity with the gcd algorithm!)

Ultimately, we reach one of the following cases:

- $\mathbf{c} = 0$ with $d > 0$. In this case, we see that $\mathrm{Res}_{d,e}(0, \mathbf{b}) = 0$.
- $\mathbf{c} = c$ is a constant and $f = 0$. In this case, we see that $\mathrm{Res}_{0,e}(c, \mathbf{b}) = c^e$.

Combining all these steps we see that there is a procedure to *calculate* the resultant *without* expanding the determinant.

Moreover, we see that if d is the degree of $\mathbf{a}$ and e is the degree of $\mathbf{b}$ the resultant $\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b})$ is 0 *if and only if* these polynomials have a common factor in $Q(Z)$. When Z is a unique factorisation domain, we have seen that this is the same the condition that they have a common factor in Z.

Splitting fields

Given an irreducible polynomial $\mathbf{p}$ of degree $d > 1$ over a field Q. We define a linear transformation α of the vector space $Q' = Q[X]_{<d}$ by

$$\{1, X, \dots, X^{d-1}\} \mapsto \{X, X^2, \dots, -(p_0 + \cdots + p_{d-1}X^{d-1})/p_d\}$$

Expressing the right-hand side in terms of the same base $\{1, X, \dots, X^{d-1}\}$ of Q' allows us to view α as a $d \times d$ matrix with entries in Q. By abuse of notation, we identify the linear transformation α with this matrix.

Sending X to α gives a ring homomorphism from $Q[X]$ to $d \times d$ matrices over Q whose image can be identified with Q' by sending X^k to α^k for $0 \leq k \leq d-1$. One easily checks that $p_0 1_d + p_1\alpha + \cdots + p_d\alpha^d$ is the 0 transformation, where 1_d denotes the identity transformation of Q'.

If $\mathbf{a}$ is a non-zero element of $Q[X]_{<d}$, then it has no common factor with $\mathbf{p}$ since the latter is irreducible and thus prime. It follows that $\gcd(\mathbf{a}, \mathbf{p})$ is an non-zero element of Q. This means that we have an identity $\mathbf{a} \cdot \mathbf{b} + \mathbf{p} \cdot \mathbf{q} = 1$ of polynomials in $Q[X]$. Substituting α in place of X we see that $\mathbf{b}(\alpha)$ gives an inverse of $\mathbf{a}(\alpha)$ in Q'.

> Given an irreducible polynomial $\mathbf{p}$ of degree $d > 1$ over a field Q, there is a field Q' containing Q which is a vector space of dimension d over Q' such that $\mathbf{p}$ has a root α in Q'.

In fact, this field is a subring of the ring of $d \times d$ matrices over Q.

Note that $\mathbf{p} = (X - \alpha)\mathbf{c}$ in $Q'[X]$ for some polynomial $\mathbf{c}$ with coefficients in Q'.

Given a polynomial $\mathbf{a}$ over Q, we can apply the above to an irreducible factor of degree greater than 1 to obtain a field Q'. We can now repeat the process for the polynomial $\mathbf{a}$ considered as a polynomial over Q'. By downward induction on the largest degree of an irreducible factor of $\mathbf{a}$, we see that there is a field K containing Q such that *all* irreducible factors of $\mathbf{a}$ are of degree 1 in $K[X]$.

> Given a polynomial $\mathbf{a}$ of degree d in $Q[X]$, there is a field K containing Q such that $\mathbf{a} = a_d \prod_{i=1}^{d}(X - \alpha_i)$ for some α_i in K.

Such a field K is called *a splitting field* for $\mathbf{a}$. (Note that since there are choices involved we are not calling it *the* splitting field.)

Application to resultants. Given a polynomial $\mathbf{a}$ of degree d in $Z[X]$, we choose a splitting field K containing $Q(Z)$ for $\mathbf{a}$, so that $\mathbf{a} = a_d \prod_{i=1}^{d}(X - \alpha_i)$ in $K[X]$. Since the resultant can be computed in *any* larger domain, we write an expression for it in K. Given a polynomial $\mathbf{b}$ of degree at most e, we claim:

$$\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) = (-1)^{de} a_d^e \prod_{i=1}^{d} \mathbf{b}(\alpha_i)$$

Note that we can put K in an even larger field L where (assuming $\mathbf{b}$ has degree e) we can write $\mathbf{b} = b_e \prod_{j=1}^{e}(X - \beta_i)$. Thus, the above expression becomes

$$\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) = a_d^e b_e^d \prod_{i=1}^{d} \prod_{j=1}^{e} (\beta_j - \alpha_i)$$

One can prove these formulas by induction on $d + e$ as follows.

Note that if $d = 0$, then $\mathbf{a} = a_0 = a_d$. As seen earlier $\mathrm{Res}_{0,e}(a_0, \mathbf{b}) = a_0^e$ which agrees with the above formula.

Hence, we can assume that $d > 0$.

Next, we note that if d is greater than the degree of $\mathbf{a}$ *and* e is greater than the degree of $\mathbf{b}$, then the above formulas do not say anything. In any case, we have seen that the resultant is 0 in this case.

By interchange, we can assume that d is equal to the degree of $\mathbf{a}$.

Now, if e is greater than the degree of $\mathbf{b}$ which is ϵ, then we have:

$$\begin{aligned}
&\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) \\
&\quad = (-1)^{d(e-\epsilon)} a_d^{e-\epsilon} \mathrm{Res}_{d,\epsilon}(\mathbf{a}, \mathbf{b}) = (-1)^{d(e-\epsilon)} a_d^{e-\epsilon} (-1)^{d\epsilon} a_d^{\epsilon} \prod_{i=1}^{d} \mathbf{b}(\alpha_i) = \\
&\qquad (-1)^{de} a_d^e \prod_{i=1}^{d} \mathbf{b}(\alpha_i)
\end{aligned}$$

where the first identity was proved earlier and the second is an application of the induction hypothesis since $d + \epsilon < d + e$.

Thus, we may assume that e equals the degree of $\mathbf{b}$.

By the interchange formula (which gives the same sign for the right-hand side of the expression above), we can also assume that $e \geq d$. This allows us use division to write $\mathbf{b} = \mathbf{q} \cdot \mathbf{a} + \mathbf{r}$ with degree ϵ of $\mathbf{r}$ *less* than d which is in turn less than e.

$$\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{b}) = \mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{r})$$

using the formula proved above. Now, we apply the above calculation to get

$$\mathrm{Res}_{d,e}(\mathbf{a}, \mathbf{r}) = (-1)^{de} a_d^e \prod_{i=1}^{d} \mathbf{r}(\alpha_i)$$

since the degree of $\mathbf{r}$ is less than e. Further, since $\mathbf{a}(\alpha_i) = 0$, we see that $\mathbf{r}(\alpha_i) = \mathbf{b}(\alpha_i)$. It follows that the right-hand side of the above formula equals $(-1)^{de} a_d^e \prod_{i=1}^{d} \mathbf{b}(\alpha_i)$ as required.

Product formula. One consequence of the above formula is

$$\mathrm{Res}_{d_1 d_2, e}(\mathbf{a}_1 \cdot \mathbf{a}_2, \mathbf{b}) = \mathrm{Res}_{d_1,e}(\mathbf{a}_1, \mathbf{b}) \mathrm{Res}_{d_2,e}(\mathbf{a}_2, \mathbf{b})$$

It would be nice to have a direct proof of this formula that does not rely on splitting fields!

Common Roots. An important consequence of the above formula is that the resultant is 0 if and only if the polynomials $\mathbf{a}$ and $\mathbf{b}$ have a common root. Of course, we have already seen that the resultant is 0 if and only if the polynomials have a common factor; so this follows from that statement also.

Discriminant. Given a polynomial $\mathbf{a}$, its *formal* derivative $D(\mathbf{a})$ is obtained by sending X^n to nX^{n-1} for $n \geq 1$, and sending constant terms to 0. It is a Z-linear map from $Z[X]$ to itself which satisfies the rules:

- $D(a) = 0$ for a in Z.
- $D(\mathbf{a} + \mathbf{b}) = D(\mathbf{a}) + D(\mathbf{b})$.
- $D(\mathbf{a} \cdot \mathbf{b}) = D(\mathbf{a}) \cdot \mathbf{b} + \mathbf{a} \cdot D(\mathbf{b})$.
- $D(X) = 1$.

In fact, D is uniquely determined by these rules.

When $\mathbf{a} = a_d \prod_{i=1}^{d}(X - \alpha_i)$ one easily checks (by induction on degree) that:

$$D(\mathbf{a}) = a_d \sum_{i=1}^{d} \prod_{j \neq i} (X - \alpha_j)$$

It follows that $D(\mathbf{a})(\alpha_i) = \prod_{j \neq i}(\alpha_i - \alpha_j)$.

The *discriminant* $\mathrm{Disc}(\mathbf{a})$ of $\mathbf{a}$ is defined as the resultant $\mathrm{Res}_{d,d-1}(\mathbf{a}, D(\mathbf{a})$.

By the above calculations, we see that if $\mathbf{a} = a_d \prod_{i=1}^{d}(X - \alpha_i)$, we have

$$\mathrm{Disc}(\mathbf{a}) = a_d^{d-1} \prod_{i=1}^{d} \prod_{j \neq i} (\alpha_i - \alpha_j)$$

Note that the discriminant is 0 if and only if $\mathbf{a}$ has repeated roots.